KMV Graph Reconstruction: Sketch-Based Approximate Adjacency for Scalable HGNN Inference

An extension of the KMV sketch propagation framework of Niu et al. [8]

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Abstract

Heterogeneous Graph Neural Network (HGNN) inference on meta-path-induced relational graphs $H_{\mathcal{P}}$ has an architectural ceiling: $H_{\mathcal{P}}$ is $O(|V|^2)$ in the worst case. On production-scale knowledge graphs — OGB MAG (736 K target nodes, 2.86–13.63 B relational edges across its three densest meta-paths) — exact materialization completes in 17–34 minutes, but the subsequent GNN forward pass is killed by out-of-memory at every depth $L \in \{2, 3, 4\}$. This paper presents **KMV graph reconstruction**, a systems-level workaround.

We extend the KMV sketch propagation framework [8] to emit a sparse approximate adjacency $\widetilde{A}$ of size $O(|V| \cdot k)$. Each hash retained in a terminal sketch points back to its origin node via the global hash map maintained at initialization; resolving these pointers reuses the propagation pass already required for centrality estimation at zero extra computation. On the full OGB MAG graph, KMV at k=8 completes both materialization (under 10 s) and inference (∼1.2 s) where Exact cannot run inference at all (Table 4).

At medium scale (ACM, DBLP, IMDB, PubMed; full materialization feasible), KMV and Meta-Path Random Walks (MPRW) [2] achieve statistically indistinguishable Macro-F1 at matched output density. MPRW retains a 1–9 pp Linear CKA advantage; we report an unexplained empirical anomaly in which MPRW retains a more homophilic edge subset than KMV at matched density on ACM and DBLP, but we do not run the importance-corrected MPRW baseline that would establish a causal mechanism. Two finite-sample bounds—a Spatial Variance Bound on per-layer aggregation and a Bounded Representation Drift on multi-layer drift—characterise KMV's intermediate behavior; neither bound implies a guarantee on downstream classification loss.

1 Introduction

Knowledge graphs (KGs) encode multi-typed entities and relationships as heterogeneous networks $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{L})$. A meta-path $\mathcal{P} = (x_0, x_1, \ldots, x_L)$ specifies a sequence of node types, and the *relational graph* $H_{\mathcal{P}} = (V, E)$ contains an edge (u, v) for every pair of target-type nodes connected by at least one meta-path instance. Heterogeneous GNNs such as HAN [9], MAGNN [4], and SeHGNN [11] require the explicit adjacency matrices of these relational graphs.

The bottleneck is *materialization*: constructing $H_{\mathcal{P}}$ costs $O(|V| \cdot |\mathcal{E}^*|)$ time and $O(|V|^2)$ space. On production-scale KGs—OGB MAG with 1.13M papers and 2.86B relational edges, OAG CS with 1.9M papers and 16.6B edges—exact materialization triggers out-of-memory failures or exceeds hour-long timeouts. Scalable GNN techniques like GraphSAGE [5], SIGN [3], and GAMLP [12] reduce training cost but still require the full adjacency matrix for pre-computation. Stochastic sampling via Meta-Path Random Walks (MPRW) [2] avoids full materialization but offers no analytical density control and yields importance-sampled (hub-biased) neighborhoods.

This paper extends KMV sketch propagation [8]—originally designed for approximate hub-query centrality on relational graphs—to act as a graph sparsifier for HGNN inference, addressing the materialization bottleneck above. Each hash retained in a terminal KMV sketch carries a pointer to its originating node via a global hash map maintained at initialization; resolving these pointers reuses the propagation pass already required for centrality estimation and produces a sparse approximate adjacency $\widetilde{A}$ of size $O(|V| \cdot k)$ at zero extra computation.

Main empirical result. On the full OGB MAG graph (736 K papers, 2.86–13.63 B edges across the three densest meta-paths), Exact materializes $H_{\mathcal{P}}$ in 17–34 minutes, but the subsequent GNN forward pass is killed by OOM at every depth $L \in \{2, 3, 4\}$. KMV at k=8 completes both materialization and inference in under ∼10 s and ∼1.2 s respectively (Table 4). This regime—where the binding constraint is downstream HGNN inference on the materialized adjacency, rather than propagation itself—is the operating point for which KMV reconstruction is intended.

Scope of claims. On medium-scale benchmarks where exact materialization is feasible, KMV does not outperform Meta-Path Random Walks (MPRW) [2] on Macro-F1; the two methods are statistically interchangeable at matched output density. The contribution of this paper is the scaling regime, not a quality advantage at small scale.

Our contributions are:

(1) **KMV Graph Reconstruction** (Algorithm 1): a simple modification to the existing centrality-oriented sketch propagation that emits a sparse approximate adjacency $\widetilde{A}$ of size $O(|V| \cdot k)$ at zero additional propagation cost.
(2) **Scaling demonstration on full OGB MAG.** Exact OOMs on inference; KMV completes in seconds (Table 4).
(3) **Two finite-sample bounds** on the propagation: Spatial Variance Bound (Lemma 4.1) on per-layer aggregation, Bounded

Code: https://github.com/gilchrisn/gnn_scalability_experiment.

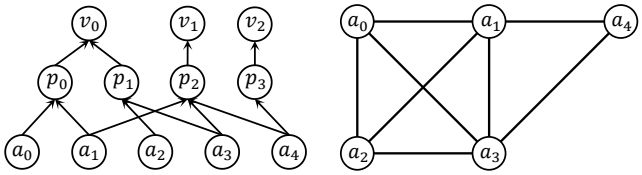

Figure 1: Left: a heterogeneous knowledge graph $\mathcal{G}$ with author, paper, and venue types. Right: the relational graph $H_{\mathcal{P}}$ induced by the meta-path Author–Paper–Venue–Paper–Author.

Table 1: Notation and key definitions used throughout the paper.

Symbol	Meaning
$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{L})$	Heterogeneous knowledge graph (typed nodes/edges)
$\mathcal{P} = (x_0, \ldots, x_L)$	Meta-path of length L over node types
$\mathcal{V}_0,\ \mathcal{V}_L$	Target-type (x_0) / terminal-type (x_L) node sets
$H_{\mathcal{P}} = (V, E)$	Relational graph induced by $\mathcal{P}$; $V = \mathcal{V}_0$
$\mathcal{N}(u)$	Meta-path neighborhood of u in $H_{\mathcal{P}}$
$\lvert\mathcal{N}(u)\rvert$	Size (cardinality) of $\mathcal{N}(u)$
$A,\ \tilde{A}$	Exact / KMV-reconstructed adjacency matrix
$r(\cdot)$	Uniform random hash $\rightarrow (0, 1)$, registered at init
$\mathcal{K}[u]$	KMV sketch at node u (size $\leq k$)
$\mathcal{H} : r \rightarrow V$	Global hash map; resolves a hash to its origin node
k	KMV sketch size (smaller $k \Rightarrow$ sparser $\tilde{A}$)
w	MPRW walk budget per source node
$L_\sigma,\ \lVert W\rVert_2$	Lipschitz constant of activation; spectral norm of weight
ε_k	Per-layer aggregation drift $O(\sqrt{d \log(d/\delta)/k})$

Representation Drift (Theorem 4.2) on multi-layer drift. Neither implies a guarantee on classification loss; their scope is stated explicitly at the top of §4.

(4) **Medium-scale evaluation.** KMV ties MPRW on Macro-F1 within seed noise on DBLP, IMDB, and PubMed; ACM is reported as a saturated meta-path control. The Linear CKA gap that favours MPRW is described, not claimed as a causal mechanism.

(5) **Structural preservation profile.** The bottom-k index $\mathcal{K}_L[v]$ preserves cardinality, expansion-decay rate (Appendix B), and Jaccard *ranking* (Appendix C); it does *not* preserve absolute Jaccard at intermediate L, nor the hub-favoured edge subset that MPRW concentrates on.

2 Background and Related Work

2.1 Knowledge Graphs and Relational Graphs

A *knowledge graph* $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{L})$ consists of typed nodes and typed edges. A *meta-path* $\mathcal{P} = (x_0, x_1, \ldots, x_L)$ is a sequence of $L + 1$ node types. A meta-path *instance* is an ordered sequence $(u_0, u_1, \ldots, u_L)$ where each u_i has type x_i and consecutive nodes are connected by an edge of the required type. The *relational graph* $H_{\mathcal{P}} = (V, E)$ has $V = \mathcal{V}_0$ (all nodes of type x_0) and $(u, v) \in E$ iff at least one meta-path instance connects them.

Figure 1 illustrates the relationship between a knowledge graph and its induced relational graph.

2.2 Notation

Table 1 summarises the symbols used throughout this paper.

Algorithm 1 KMV Graph Reconstruction

Require: KG $\mathcal{G}$, meta-path $\mathcal{P}$, sketch size k
Ensure: Sparse approximate adjacency $\tilde{A}$

```
Ã ← 0; H ← ∅
for u ∈ V_0 do
    r(u) ← Rand(0, 1)
    H[r(u)] ← u; K[u] ← {r(u)}
end for
PROPAGATE(K) along P          // base sketch propagation [8]
for v ∈ V_L do
    u_v ← node in V_0 corresponding to v
    for x ∈ K[v̄] do
        u_x ← H[x]                                  {O(1) lookup}
        Ã[u_x, u_v] ← 1; Ã[u_v, u_x] ← 1
    end for
end for
return Ã
```

2.3 KMV Sketch Propagation

The K-Minimum Values (KMV) sketch [1] estimates set cardinality by retaining the k smallest uniformly random hash values. Given a set C, assign each element o a hash $r(o) \sim \text{Uniform}(0, 1)$, and define $\mathcal{K}(C) = \{k \text{ smallest } r(o) : o \in C\}$. The cardinality estimator is $\lvert\widetilde{C}\rvert = (k - 1)/\max \mathcal{K}(C)$.

The base paper [8] introduces *sketch propagation*: initialize each target-type node with its own hash, then propagate sketches along meta-path edges, merging at each level by retaining the k smallest values. Terminal sketches enable hub-query centrality estimation (degree, h-index) with > 90% F1 and 10–191× speedup over exact methods.

2.4 Related Work

HGNNs. HAN [9], MAGNN [4], SeHGNN [11], and HeCo [10] all assume pre-materialized adjacency matrices.

Scalable GNNs. GraphSAGE [5], SIGN [3], and GAMLP [12] reduce training cost but still require the full adjacency for precomputation.

Stochastic sampling. MPRW / Metapath2Vec [2] avoids full materialization via random walks but provides importance-sampled (hub-biased) neighborhoods with no analytical density control.

Gap. No existing method provides memory-bounded approximate adjacency with formal guarantees for HGNN inference on unmaterialized relational graphs.

3 KMV Graph Reconstruction

3.1 Algorithm

The key observation is that a global hash map $\mathcal{H} : r(u) \rightarrow u$ registered at initialization allows each terminal sketch hash to be resolved to its originating node in $O(1)$. This converts the statistical KMV data structure into a topological one at zero extra propagation cost.

3.2 Complexity

- **Propagation**: $O(|\mathcal{E}^*| \cdot k)$ — each edge traversed once per sketch entry.
- **Reconstruction**: $O(|V| \cdot k)$ hash lookups.
- **Space**: $O(|V| \cdot k)$ for sketch arrays and output $\tilde{A}$.

Compared to exact materialization ($O(|V|^2)$ space), KMV reconstruction reduces memory by a factor of $|V|/k$. For ACM ($|V| = 3{,}025$, $k = 32$): exact adjacency has 9.15M edges; KMV produces 96.8K edges — a 94.5× reduction.

3.3 Inference Complexity

Exact GNN message-passing requires $O(L \cdot |V|^2 \cdot d^2)$ per layer (dense A). With KMV's sparse $\tilde{A}$ ($\leq |V| \cdot k$ non-zeros), inference cost becomes $O(L \cdot |V| \cdot k \cdot d^2)$ — converting the quadratic bottleneck to linear.

4 Theoretical Analysis

Scope of the bounds in this section. Lemma 4.1 bounds the per-layer drift of KMV's feature aggregation $\tilde{\mu}_u$ against the exact aggregation μ_u, in ℓ_∞ on a single coordinate. Theorem 4.2 composes per-layer drift through L Lipschitz layers to bound the representation drift $\|\tilde{h}_u^{(L)} - h_u^{(L)}\|_2$. *Neither bound implies a guarantee on classification loss or on* arg max *agreement with the true label* y. A bounded representation drift can still flip a arg max at the readout layer, and conversely a representation can change substantially without changing arg max on the held-out test set. We therefore use the bounds as predictors of *intermediate* faithfulness (depth-wise CKA, Pred. Agreement against an Exact-trained reference embedding), not of downstream task performance, which we measure separately and empirically in §6.

4.1 Spatial Variance Bound

Lemma 4.1 (Spatial Variance Bound). *Let $x_v \in \mathbb{R}^d$ with $\|x_v\|_\infty \leq \mathcal{M}$. Let $\tilde{\mu}_{u,j}$ and $\mu_{u,j}$ be the KMV-approximated and exact feature aggregation for dimension j. If $|\mathcal{N}(u)| > k$, then by Bernstein's inequality for sampling without replacement:*

$$\Pr\big(|\tilde{\mu}_{u,j} - \mu_{u,j}| \geq t\big) \leq 2\exp\left(-\frac{k\,t^2}{2\sigma_j^2 + \frac{2}{3}\mathcal{M}\,t}\right)$$

where $\sigma_j^2 = \mathrm{Var}(x_{v,j})$ over $v \in \mathcal{N}(u)$.

Key properties: (i) $\mathbb{E}[\tilde{\mu}_u] = \mu_u$ (unbiased), (ii) error decays as $O(1/\sqrt{k})$, (iii) **independent of $|V|$ and Λ (max degree)**. When $|\mathcal{N}(u)| \leq k$, the full neighborhood is retained exactly.

4.2 Bounded Representation Drift

Theorem 4.2 (Bounded Representation Drift). *Let an HGNN apply weight $W^{(\ell)}$ and L_σ-Lipschitz activation σ at each layer $\ell = 1, \ldots, L$. Define $C^{(\ell)} = L_\sigma \cdot \|W^{(\ell)}\|_2$ and $C = \max_\ell C^{(\ell)}$. Then the representation drift at depth L satisfies:*

$$\|\tilde{h}_u^{(L)} - h_u^{(L)}\|_2 \leq \varepsilon_k \cdot L \cdot C^L$$

where $\varepsilon_k = O\left(\sqrt{d\log(d/\delta)/k}\right)$ with probability $\geq 1 - \delta$.

Proof sketch. At a single layer: $\|\tilde{h}_u - h_u\|_2 \leq L_\sigma \cdot \|W\|_2 \cdot \|\tilde{\mu}_u - \mu_u\|_2$ by Lipschitz continuity. Applying Lemma 4.1 with threshold $t = \sqrt{(c/k)\log(2d/\delta)}$ and a union bound over d dimensions: $\|\tilde{\mu}_u - \mu_u\|_2 \leq \varepsilon_k$. Multi-layer drift decomposes recursively: $\Delta^{(\ell+1)} \leq C^{(\ell)} \cdot \Delta^{(\ell)} + C^{(\ell)} \cdot \varepsilon_k$. Unrolling from $\Delta^{(0)} = 0$ yields the result. □

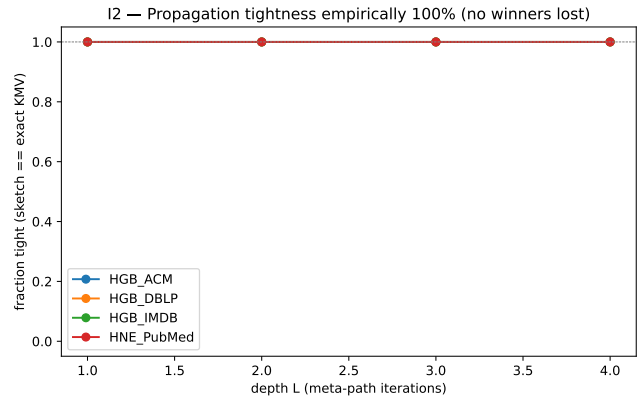

Figure 2: KMV propagation tightness: the fraction of (node, L) combinations on which the bottom-k-of-bottom-k propagation introduces zero leak relative to bottom-k of the full L-hop neighborhood. Across four datasets and $L \in \{1, 2, 3, 4\}$, the empirical leak rate is 0.

The bound decays with k (more sketch budget → tighter approximation) but grows with L (deeper networks compound per-layer error)—the *budget–depth trade-off* measured empirically in Section 6.

4.3 Empirical Tightness

The exponential-in-L factor C^L in Theorem 4.2 is an artefact of the recursive Lipschitz unrolling. Empirically the propagation itself is exact: the bottom-k-of-bottom-k merge introduces zero leak relative to bottom-k of the full L-hop neighborhood (Figure 2). Across four datasets and $L \in \{1, 2, 3, 4\}$ the leak rate is identically 0. Theorem 4.2's exponential-in-L drift bound is therefore loose by construction; it bounds the proof technique, not the algorithm in practice.

5 Experimental Methodology

5.1 Two-Stage Inductive Protocol

Training a graph encoder directly on a stochastic sparsifier's output $\tilde{A}$ exposes the gradient to edge-set noise that varies between runs and between epochs, destabilising optimisation. We therefore adopt a two-stage inductive protocol that decouples the sparsifier from training. Stage I uses the exact adjacency to fit encoder weights; Stage II uses the sparsifier at inference only, with frozen weights.

Stage I (training). Partition the target-type node set $\mathcal{V}_0$ into a training subset V_{train} ($|V_{\text{train}}|/|\mathcal{V}_0| = 0.4$) and a test subset V_{test} (the remaining 0.6). Where the underlying KG carries temporal metadata on the target type — e.g., paper publication year for DBLP and OGB MAG — the partition is chronological, with V_{train} containing the oldest 40%. Otherwise the partition is uniform at random with seed 42. Materialise the exact relational graph A_{train}, restricted to the subgraph induced by V_{train} (edges (u, v) where both $u, v \in V_{\text{train}}$).

Train a GraphSAGE encoder f_θ with mean aggregation, hidden dimension 64, on $(A_{\text{train}}, X_{V_{\text{train}}})$ to convergence: cross-entropy loss, Adam optimiser, early stopping with patience 30, maximum 200 epochs. Save the frozen weights θ^*. We repeat this for each model depth $L \in \{1, 2, 3, 4\}$ separately, producing four checkpoints per dataset.

Stage II (inference). For each method under test, apply the corresponding materialiser to the *full* graph (i.e. to $\mathcal{V}_0$, not to V_{train}) to obtain a target adjacency $\widetilde{A}$. Compute $Z = f_{\theta^*}(\widetilde{A}, X)$ followed by the linear classifier head co-trained in Stage I. All metrics are computed on V_{test} only — nodes the encoder has not seen edges incident to during training.

5.2 Baselines

We compare three materialisers, each invoked through a uniform child-process interface for measurement parity (§5.4).

Exact. The exact relational graph $A = H_{\mathcal{P}}$, computed by the C++ implementation released with the base paper [8], which we use unmodified. When Exact is used at inference, Z_{exact} is the reference embedding against which the two approximators are measured.

KMV. Algorithm 1, implemented in C++ as an additional command on top of the same propagation engine used for Exact. The output $\widetilde{A}_{\text{KMV}}$ has at most $|V| \cdot k$ entries.

MPRW (Meta-Path Random Walks) [2]. For each $u \in \mathcal{V}_0$, launch w independent random walks following meta-path $\mathcal{P}$. The retained adjacency $\widetilde{A}_{\text{MPRW}}$ is the union of (source, terminal) pairs over all completed walks. MPRW runs as a w-sweep with each w a separate materialisation; we do not perform density-matching calibration because the resulting density is part of the comparison axis (§5.3). MPRW is implemented as a native C++ binary using the same graph-loading and timing infrastructure as Exact and KMV; this avoids interpreter overhead in the memory measurement.

5.3 Parameter Grid

The empirical grid producing the figures of §6 and the data of Table 3 sweeps three parameters:

$k \in \{2, 4, 8, 16, 32, 64\}$	(KMV sketch size)
$w \in \{1, 2, 4, 8, 16, 32, 64, 128, 256, 512\}$	(MPRW walk budget)
$L \in \{1, 2, 3, 4\}$	(model depth)

KMV k and MPRW w both range over consecutive powers of two; the w-range extends one decade beyond the k-range so MPRW reaches densities matched to KMV at the larger k values.

Replicates. For every $(k, L, \text{dataset})$ cell we run 50 KMV replicates with independent hash seeds. For every $(w, L, \text{dataset})$ cell we run 10 MPRW replicates with independent walk-RNG seeds (eight on IMDB). Exact is deterministic; we record one run per $(L, \text{dataset})$ cell. Cell-level standard deviations reported in Table 3 are computed across the replicates of that cell. Cross-dataset comparisons fix k=32, w=32, L=2 as the operating point unless stated otherwise.

5.4 Measurement Protocol

We require comparable timing and memory measurements across Exact, KMV, and MPRW. All three materialisers are invoked as child processes under a uniform external wrapper that records peak resident memory and wall-clock time, ensuring that the reported metrics come from a single, instrument-agnostic source.

Peak resident memory is read from the wrapper's report. *Algorithm wall-clock* is read from a timing line emitted by the C++ implementation itself; the timer is started *after* the typed graph G^* is loaded into memory and stopped before output is written. All three implementations emit this line at the same point relative to graph load, so the reported time excludes graph load uniformly. *Inference* is measured by the same wrapper: the frozen θ^* checkpoint is loaded, the materialiser's output is fed through, and inference time and peak memory are recorded. An entry of *INF-FAIL* in Table 4 denotes a process killed by the operating system during inference for exceeding available memory; the materialisation step in such rows completed normally.

5.5 Fidelity Metrics

Every metric below is computed on V_{test} only.

Macro-F1. Standard multi-class macro-averaged F1 of the linear classifier head's predictions against the ground-truth labels of V_{test}. Reported as the mean over replicates with cell-level standard deviation.

Linear CKA [7]. For two embedding matrices $Z_{\text{exact}}, Z_{\text{approx}} \in \mathbb{R}^{|V_{\text{test}}| \times d}$, $\text{CKA}(Z_{\text{exact}}, Z_{\text{approx}}) = \frac{\|Z_{\text{approx}}^\top Z_{\text{exact}}\|_F^2}{\|Z_{\text{exact}}^\top Z_{\text{exact}}\|_F \cdot \|Z_{\text{approx}}^\top Z_{\text{approx}}\|_F}$ (centered features). CKA is invariant to orthogonal transforms and isotropic scaling — the standard representation-similarity metric for comparing sparsifier outputs to a reference encoder.

Prediction Agreement (PA). The fraction of V_{test} on which $\arg\max Z_{\text{exact}}(u) = \arg\max Z_{\text{approx}}(u)$ at the linear classifier head. This is the 0/1 classification analogue of CKA — a direct measure of how often the approximate model would make the same decision as the exact one.

Depth-wise CKA. For an L-layer encoder, the per-layer Linear CKA between exact and approximate intermediate representations $h_{\text{exact}}^{(\ell)}$ and $h_{\text{approx}}^{(\ell)}$ at each $\ell \in \{1, \ldots, L\}$. Used to localise where representational drift accrues with depth.

Dirichlet Energy. For the encoder output $h \in \mathbb{R}^{|V| \times d}$, the standard graph-signal Dirichlet energy $\mathcal{E}(h) = \sum_{(u,v) \in E} \|h_u - h_v\|_2^2$ computed against the exact adjacency $E = E(A_{\text{exact}})$. A diagnostic for over-smoothing collapse as depth increases.

We use Macro-F1 against the ground-truth label as the headline downstream metric and Linear CKA / PA against Z_{exact} as the headline fidelity metrics. The drift bounds of §4 apply to intermediate aggregations and are therefore most directly tested by depth-wise CKA, not by Macro-F1.

6 Empirical Results

6.1 Setup

We follow the methodology of §5: the two-stage protocol of §5.1, the three baselines of §5.2, the parameter grid of §5.3, the measurement protocol of §5.4, and the fidelity metrics of §5.5. This subsection records dataset-specific quantities not part of the methodology.

Datasets. We evaluate on four heterogeneous knowledge graphs (Table 2). On each dataset we use the densest meta-path that yields

Table 2: Dataset statistics. $|E_{\text{exact}}|$ is the exact directed-edge count of the relational graph $H_{\mathcal{P}}$ (entries of A including self-loops). Density $\rho = |E_{\text{exact}}|/|V|^2$. ACM PTP and DBLP APTPA are saturated meta-paths ($\rho \to 1$): every paper-paper or author-author pair has at least one shared term.

Dataset	Meta-path	$\lvert V\rvert$	$\lvert E_{\text{exact}}\rvert$	ρ
ACM	P→T→P	3,025	9.15 M	1.00
DBLP	A→P→T→P→A	4,057	16.46 M	1.00
IMDB	M→K→M	4,763	10.15 M	0.45
PubMed	D→C→D	12,244	72.79 M	0.49

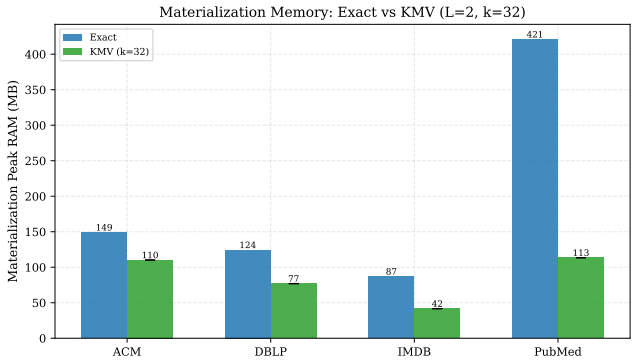

Figure 3: Peak materialization RAM: KMV (k=32) vs Exact. KMV is bounded by $O(|V| \cdot k)$; Exact by $O(|V|^2)$.

a target-type relational graph; the densest meta-path is the adversarial choice for any sparsifier because the exact $H_{\mathcal{P}}$ approaches saturation ($\rho \to 1$).

Operating point. Cross-dataset comparisons in Table 3 fix the operating point at L=2, k=32 for KMV and w=32 for MPRW; full k- and L-sweeps in Figs. 5–6 cover the parameter grid of §5.3. Layer-wise CKA trajectories are reported in Appendix A.

6.2 Systems Efficiency

Figure 3 shows peak materialization RAM. KMV at k=32 uses 42–113 MB across all datasets, while Exact requires 124–421 MB. The gap is most pronounced on PubMed (113 vs 421 MB, 3.7×), where the dense relational graph (72.8M edges) approaches $O(|V|^2)$.

Figure 4 shows materialization time. KMV completes in 0.08–0.50 s; Exact requires 4.1–14.3 s (29–64× slower). KMV's single deterministic pass has no calibration overhead — a single analytical knob k controls density.

6.3 Representational Fidelity

Table 3 presents downstream quality at L=2, k=32.

We exclude ACM from the cross-dataset aggregate because of meta-path saturation (see the saturation control discussion below). On the three non-saturated benchmarks (DBLP, IMDB, PubMed) at matched density (k=32, w=32), KMV and MPRW Macro-F1 are within ∼1 pp of each other. On DBLP the three methods are within 0.7 pp (KMV 0.763, MPRW 0.767, Exact 0.769). On PubMed they are

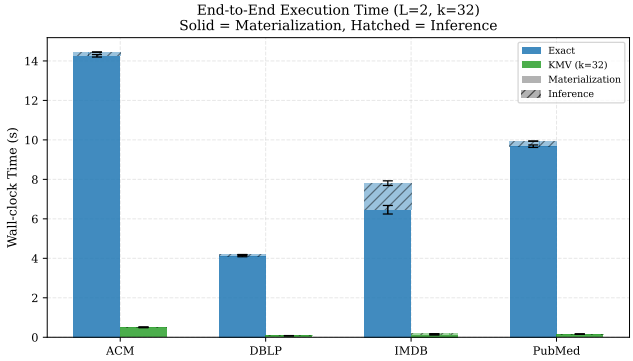

Figure 4: Materialization time: KMV (k=32) vs Exact. KMV is a single deterministic pass; Exact requires full graph construction.

Table 3: Downstream fidelity at L=2, k=32. MPRW shown at natural w=32 operating point. Metrics on V_{test} only.

Dataset	Method	Exact F1	Macro-F1	CKA	Pred.Agr.
ACM	KMV	0.813	$0.721_{\pm.001}$	0.927	$0.846_{\pm.001}$
	MPRW		$0.807_{\pm.003}$	0.998	$0.976_{\pm.004}$
DBLP	KMV	0.769	$0.763_{\pm.002}$	0.976	$0.962_{\pm.002}$
	MPRW		$0.767_{\pm.003}$	0.993	$0.958_{\pm.002}$
IMDB	KMV	0.520	$0.565_{\pm.002}$	0.889	$0.872_{\pm.004}$
	MPRW		$0.571_{\pm.001}$	0.925	$0.898_{\pm.003}$
PubMed	KMV	0.306	$0.318_{\pm.002}$	0.904	$0.937_{\pm.001}$
	MPRW		$0.312_{\pm.004}$	0.979	$0.977_{\pm.001}$

within 1.2 pp (KMV 0.318, MPRW 0.312, Exact 0.306). On IMDB both sparsifiers exceed Exact by ∼5 pp (KMV 0.565, MPRW 0.571, Exact 0.520); the offset holds for both, so we do not claim a regularisation effect.

CKA consistently favours MPRW: 0.93–1.00 vs KMV's 0.89–0.98, a 1–9 pp gap across the four datasets. Higher CKA is *not* aligned with higher F1: on DBLP MPRW's 0.993 CKA pairs with 0.767 F1 vs KMV's 0.976/0.763—the F1 gap is 0.4 pp despite a 1.7 pp CKA gap. MPRW maintains a 1–9 pp Linear CKA advantage over KMV. We observe an unexplained empirical anomaly where MPRW retains a significantly more homophilic edge subset than KMV at matched densities (see Appendix D); establishing whether this is the causal mechanism for the CKA gap remains open.

ACM as a saturation control. ACM PTP at ρ=1.00 is saturated: every paper-paper pair shares at least one term, so $H_{\mathcal{P}}$ is a near-complete graph. KMV F1 (0.721) drops ∼9 pp below Exact (0.813) and MPRW (0.807) on this dataset; the gap reflects KMV's edge-count budget hitting a topological ceiling specific to the saturation regime, rather than a representative comparison. We retain ACM in Table 3 as a control on what happens when a sparsifier is applied to a graph that is already maximally connected, and exclude it from cross-dataset aggregates.

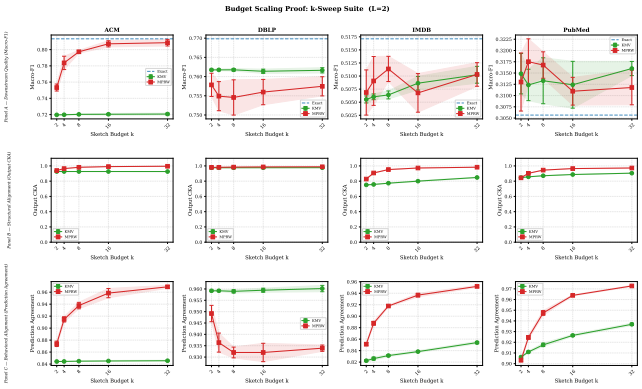

Figure 5: Quality vs sketch budget k at L=2. Top: Macro-F1. Middle: Output CKA. Bottom: Prediction Agreement. Exact shown as horizontal ceiling; MPRW as horizontal band at fixed w=32.

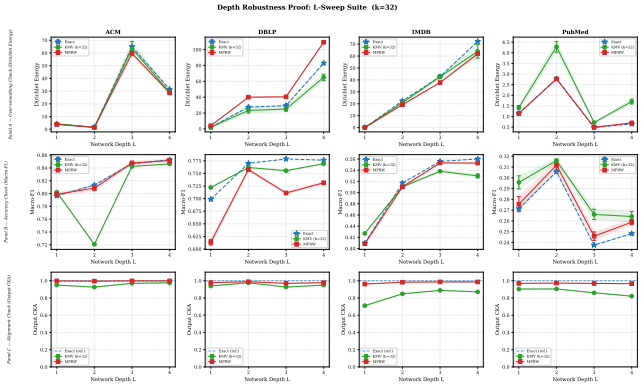

Figure 6: Quality vs depth L at k=32. Top: Dirichlet Energy. Middle: Macro-F1. Bottom: Output CKA. MPRW at fixed w=32.

6.4 Budget Scaling (k-sweep)

Figure 5 shows all three quality metrics as a function of k over the plotted range $k \in [2, 32]$. On ACM and DBLP all three metrics are essentially flat across this range (Macro-F1 changes by less than 0.5 pp on both): the saturated meta-paths (ρ=1) reach a topological ceiling already at k=2. On IMDB and PubMed, CKA and Prediction Agreement improve monotonically with k and are still climbing at k=32 (PubMed CKA: 0.85 $\rightarrow$ 0.92 from k=2 to k=32; IMDB CKA: 0.73 $\rightarrow$ 0.91); the $O(1/\sqrt{k})$ bound from Lemma 4.1 is visible on these structural metrics. Macro-F1 is roughly flat across k on every dataset (within 1 pp), with a mild non-monotonicity on IMDB—consistent with SAGE-mean's robustness to neighborhood identity at fixed cardinality.

6.5 Depth Robustness (L-sweep)

Figure 6 shows depth robustness. KMV CKA degrades progressively with L (PubMed: 0.904 $\rightarrow$ 0.821 from L=1 to L=4), consistent with the $\varepsilon_k \cdot L \cdot C^L$ prediction of Theorem 4.2. MPRW CKA stays near 0.98 at all depths (PubMed: 0.976 $\rightarrow$ 0.977), reflecting structurally coherent walk sampling.

The Dirichlet Energy panel reveals over-smoothing dynamics. On DBLP at L=4, MPRW's Dirichlet Energy overshoots Exact (~109 vs ~83) — indicating over-differentiation from stochastic walk noise — without corresponding task-loss improvement.

Table 4: Scalability on the *full* OGB MAG graph (736,389 target-type papers) across three meta-paths spanning 2.86 B–13.63 B edges. Exact materializes the relational graph (17–34 minutes, 23–65 GB resident) but the subsequent GNN forward pass is killed by OOM at every depth $L \in \{2, 3, 4\}$. KMV at k=8 completes both materialization *and* inference. Numbers are means over $L \in \{2, 3, 4\}$.

Meta-path	Method	$\lvert\widetilde{A}\rvert$	Mat (s)	Mat RAM	Inference
rev_writes,writes	Exact	2.86 B	1021	23.3 GB	**INF-FAIL**
	KMV k=8	5.24 M	9.9	8.6 GB	1.2 s, OK
cites,rev_cites	Exact	13.63 B	2026	64.9 GB	**INF-FAIL**
	KMV k=8	4.63 M	8.7	8.4 GB	1.2 s, OK
rev_cites,cites	Exact	7.83 B	1431	41.3 GB	**INF-FAIL**
	KMV k=8	4.64 M	8.8	8.4 GB	1.2 s, OK

6.6 Large-Scale Scalability

Table 4 demonstrates the regime KMV is designed for: at the full OGB MAG [6] scale, the binding constraint is not propagation but the *downstream* inference pass on the materialized adjacency. Exact's meta-path adjacencies span 2.86 B to 13.63 B edges; constructing them succeeds in 17–34 minutes, but the subsequent GNN forward pass exceeds the available memory and is killed by the OS at every depth. KMV at k=8 produces a 4–5 M-edge approximate adjacency in under 10 s, runs inference in ~1.2 s, and uses ~8.5 GB throughout. The smaller and larger budgets behave consistently: k=2 produces ~1.3 M-edge adjacencies in ~7 s, and k=32 produces ~15 M-edge adjacencies in ~25 s; all complete inference within ~1.3 s. On this scale, Exact materialises $H_{\mathcal{P}}$ but cannot run inference; KMV completes both stages in seconds.

7 Discussion

7.1 KMV vs MPRW: Operational Comparison

The experimental evidence shows that KMV and MPRW deliver *comparable* downstream quality. Neither method dominates across all datasets: MPRW leads on ACM PTP at L=2 (the saturated, hub-dense regime), and the two are statistically interchangeable on DBLP, IMDB, and PubMed.

The contribution of KMV is therefore not a quality advantage over MPRW but a set of operational properties as an approximation of Exact:

- **Space.** $O(|V| \cdot k)$ output, in contrast to Exact's $O(|V|^2)$.
- **Density control.** A single analytical parameter k controls the output edge count; no calibration sweep is required.
- **Determinism.** A single deterministic propagation pass; no seed variance budget.
- **Finite-sample bounds.** Lemma 4.1 and Theorem 4.2 characterise per-layer and multi-layer aggregation drift, in contrast to MPRW's asymptotic convergence in w.

Table 5: KMV vs MPRW: operational comparison.

Property	KMV	MPRW
Sampling	Uniform (hash)	Importance (walks)
Density ctrl	Analytical (k)	Emergent (w)
Passes	1 deterministic	w per node
Bias	Label-blind	Homophily-skewed (§D)
Guarantees	Finite-sample (k)	Asymptotic ($w \to \infty$)

MPRW's higher CKA (0.973–0.996 vs. 0.848–0.977) reflects its importance-sampled walk structure preserving the hub geometry of the exact graph (Appendix D). However, this alignment is concentrated in a class-agnostic hub subspace, and higher CKA does not translate to higher F1 on three of the four medium-scale datasets (Table 3).

7.2 What $\mathcal{K}_L[v]$ Preserves

The interchangeability of KMV and MPRW for the SAGE-mean classifier is a property of the downstream task, not a deficiency of either sparsifier. Mean-aggregating GraphSAGE on a sparsified adjacency depends only on the *cardinality* of each retained neighborhood, not on *which* neighbors are retained: at matched cardinality, the two sparsifiers are interchangeable substrates. The structural information the bottom-k index $\mathcal{K}_L[v]$ does carry—each retained hash points back to a specific origin node via the global hash map $\mathcal{H}$, so any two nodes' neighborhoods are not just sized but *identifiable*, and hence comparable—is unused under mean aggregation.

The experiments in this paper supply a partial profile of which structural properties of the exact L-hop neighborhood the index $\mathcal{K}_L[v]$ preserves and which it loses. On the preservation side: *cardinality* (the original purpose of the sketch [8]); *per-node expansion-decay rate*, with $\log \max \mathcal{K}_L[v]$ slope correlating with $\log |N_L(v)|$ slope at Pearson $r = -0.997$ on PubMed, -0.992 on IMDB, and -0.789 on DBLP (Appendix B); *bottom-k Jaccard ranking* between L-hop neighborhoods at Pearson $r > 0.93$ across $L \in \{1, \ldots, 4\}$ on three datasets (Appendix C); and—against an MPRW baseline at matched density—a 2–6× higher spectral radius $\lambda_{\max}$, evidence that KMV preserves expander structure where MPRW collapses to a walk skeleton (Appendix D). On the loss side: *absolute Jaccard at intermediate L* is biased downward by up to 10 pp—the ranking is preserved but absolute calibration is not (Appendix C); the *hub-favoured edge subset* on which MPRW concentrates is not retained, which is a disadvantage for KMV on homophilic classification tasks where the discriminative structure lives in the hub subgraph (the ACM saturation case in Table 3 is the extreme); and arg max *agreement on downstream classification* is not implied by the drift bounds of §4, which apply to intermediate aggregations only.

7.3 Limitations

We list the threats to the claims of this paper explicitly.

- **Bounds do not cover classification loss.** Lemma 4.1 bounds the per-layer aggregation drift, and Theorem 4.2 bounds the representation drift. Neither bounds the classification loss or the agreement with the true label y.
- **The MPRW baseline is uncorrected importance sampling.** An importance-reweighted MPRW baseline [3, 12] that removes the homophily bias measured in §D is not included in this evaluation; the residual MPRW-vs-KMV CKA gap may shrink under such a correction.
- **The HGNN architecture used is GraphSAGE, not Simple-HGN.** Our quality numbers are GraphSAGE Macro-F1; HGB benchmark results suggest tuned simple architectures can establish a stronger ceiling.
- **Saturation regimes are degenerate.** ACM PTP at ρ=1.00 (§6.3) and ACM I4 (Appendix B) reflect a saturated metapath where the relational graph is a near-clique; the fidelity numbers and the slope correlation respectively become uninformative there.
- **Memory advantage is at the output, not the working set.** On the C++ binary used for measurement, KMV and Exact have comparable in-process RAM at the medium-scale benchmarks; the savings appear in the output adjacency size and at the inference stage on graphs where the output dominates.

7.4 Future Work

- **Tighten Theorem 4.2.** Empirically the propagation is tight: no information is lost in the bottom-k-of-bottom-k merge across the four datasets and $L \in \{1, \ldots, 4\}$ (§4.3). The current exponential-in-L bound is therefore loose by construction. A topology-aware tighter bound conditioning on per-node max-intermediate-degree is open.
- **Importance-corrected MPRW.** A reweighted MPRW baseline (§D) is the clean test of whether the homophily structure of MPRW's retained edges is causally responsible for the CKA gap.
- **Motif and spectral preservation.** The bottom-k representation preserves *set* information but not *multiset*. Whether L-hop motif counts and spectral statistics survive the propagation is open, and would identify which classes of downstream task can consume $\mathcal{K}_L$ directly.
- **Sketch-aware training.** End-to-end training on $\widetilde{A}_{\text{KMV}}$ with variance reduction (control variates, antithetic sampling), removing the requirement that an Exact training subgraph be feasible at all.
- **Streaming and temporal updates.** KMV propagation is monotone (per-level min only decreases); sketches admit incremental updates as edges arrive. Quantifying the regret of incremental KMV against periodic re-propagation is open.
- **Stronger HGNN backbone.** The 1 pp KMV/MPRW/Exact tie at medium scale is reported under GraphSAGE with mean aggregation. A tuned Simple-HGN-style architecture would establish a stronger Macro-F1 ceiling and is the natural next quality benchmark.

8 Conclusion

This paper presents KMV graph reconstruction (Algorithm 1), a systems-level workaround for the $O(|V|^2)$ ceiling of exact relational-graph materialization in HGNN inference. The main empirical result

is that on the full OGB MAG graph (three densest meta-paths, 2.86–13.63 B edges) Exact's GNN forward pass is killed by OOM at every depth, while KMV at k=8 completes both materialization and inference in seconds (Table 4). Two finite-sample bounds (Lemma 4.1, Theorem 4.2) characterise KMV's intermediate-feature behavior; their scope is stated explicitly in §4 and does not extend to classification loss.

At medium scale, KMV and MPRW are statistically interchangeable on Macro-F1 at matched density (within ∼1 pp on DBLP, IMDB, and PubMed; ACM is reported as a saturation control). MPRW retains a Linear CKA advantage which we describe but do not establish as causal; the importance-corrected MPRW baseline that would test the homophily-bias hypothesis is listed as open work (§7.4). The mechanistic question of which structural properties of $H_{\mathcal{P}}$ the bottom-k index $\mathcal{K}_L[v]$ preserves—cardinality, expansion-decay rate, and rank-Jaccard yes; absolute Jaccard and the hub-favoured edge subset no—is the program we hand to follow-on work.

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A Layer-wise CKA Trajectory

This appendix serves as the empirical validation of the drift bound in Theorem 4.2.

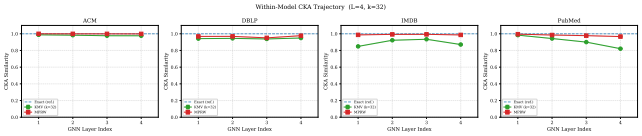

Figure 7: Layer-wise CKA trajectory within a depth-4 model (k=32). Each point ℓ shows CKA between the Exact-trained reference and the approximate model at intermediate layer ℓ.

Figure 7 shows how representational alignment evolves within a depth-4 model. MPRW remains nearly flat near 1.0 across all layers. KMV shows a steeper decline, consistent with the recursive drift expansion in Theorem 4.2; the degradation follows the analytical decay curve of §4 rather than collapsing.

B Expansion-Decay Signal

This is a secondary structural property of the sketch and is not theoretically novel (bottom-k max-hash scaling is implicit in the HIP estimator literature); we report it as an empirical consequence the same sketch already exposes.

The KMV sketch carries an additional signal beyond average aggregation: the rate at which $\log \max \mathcal{K}_L[v]$ shrinks with depth L. Under a per-node expansion model $|N_L(v)| \approx |N_{L-1}(v)| \cdot \beta_v$, the expectation of $\max \mathcal{K}_L[v]$ scales as $k/|N_L(v)|$, giving the relation

$$\mathrm{slope}_L\big(\log \max \mathcal{K}_L[v]\big) \;=\; -\,\mathrm{slope}_L\big(\log |N_L(v)|\big).$$

Table 6: Empirical agreement between per-node max-hash slope and ground-truth log**-cardinality slope across $L \in \{1, 2, 3, 4\}$ at k=64. Pearson r between the two slopes is the metric of interest. ACM saturates to the full target set at L=1, so its slope is degenerate by construction.**

Dataset	r(slope-max, slope-card)	p-value	n
HGB_ACM	saturates at L=1	—	300
HGB_DBLP	−0.789	2.4×10^{-15}	67
HGB_IMDB	−0.992	$< 10^{-262}$	290
HNE_PubMed	−0.997	1.1×10^{-125}	110

Table 6 reports per-node Pearson correlations between the max-hash slope and the ground-truth cardinality slope. The agreement is essentially perfect on PubMed and IMDB ($r = -0.997$, -0.992) and strong on DBLP ($r = -0.789$). The signal is recoverable from the sketch with no extra propagation pass and no additional memory beyond the sketch itself, enabling adaptive-depth message passing, structural-role detection via slope clustering, and online mixing-time estimation at $O(|V| \cdot k)$ memory.

C Bottom-k Jaccard Preservation

The bottom-k representation preserves Jaccard *rank* between neighborhoods but fails to preserve absolute Jaccard at intermediate depths—an explicit limitation of the sketch as a similarity estimator.

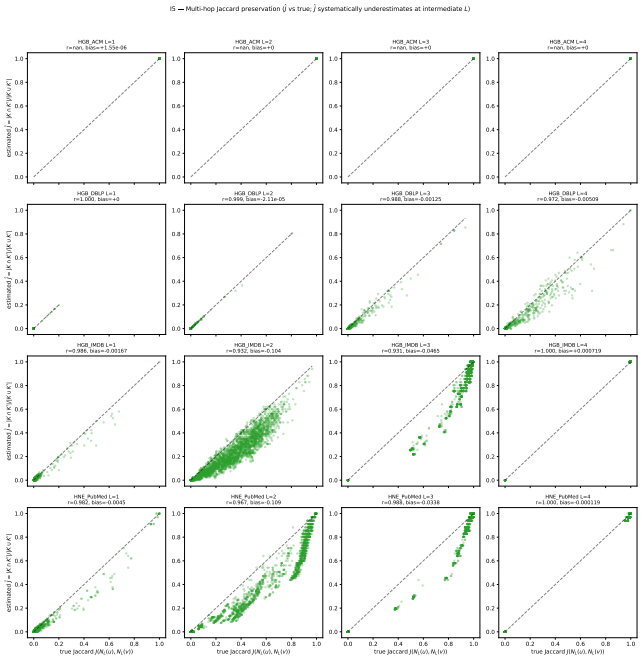

Figure 8: Estimated vs. ground-truth Jaccard between L-hop neighborhoods, sampled per pair across $L \in \{1, 2, 3, 4\}$ on three datasets. Pearson rank-correlation exceeds 0.93 throughout, but the absolute estimate has a systematic negative bias peaking at L=2 (−10 pp on IMDB and PubMed) and recovering at L=4 as neighborhoods saturate.

The bias is a calibration issue, not a ranking one. Downstream consumers that depend on the *ordering* of neighborhood-pair similarities (e.g., link prediction, nearest-neighbor retrieval) inherit a near-faithful signal; consumers that depend on absolute-Jaccard magnitudes do not.

D Structural Anomalies: Homophily Bias and Spectral Radius

This appendix records two empirical structural differences between the KMV-retained and MPRW-retained edge sets at matched density. We frame both as *unconfirmed empirical findings*: they are unresolved artefacts of importance-sampled walk completion versus uniform-hash sampling, and we do not establish a causal link to the downstream CKA gap reported in §6.3.

Edge homophily. For an undirected sparsified adjacency $\widetilde{A}$ and a class label y_u assigned to each target node u,

$$\text{homophily}(\widetilde{A}) = \frac{|\{(u,v) \in \widetilde{A} : y_u = y_v\}|}{|\{(u,v) \in \widetilde{A} : y_u, y_v \text{ both labelled}\}|},$$

i.e. the fraction of retained edges whose two endpoints carry the same class label. ACM and DBLP have full label coverage on their target node sets. PubMed has 1.83% label coverage and is statistically underpowered for this measurement; we exclude it from the headline numbers.

Table 7: Edge homophily of the retained adjacency at matched output density on ACM and DBLP. Densities are the sparsifier's own (KMV's edge count vs. Exact's $|E_{\text{exact}}|$ on the left, MPRW's on the right). Δ = MPRW − KMV.

Dataset	KMV dens.	KMV homo.	MPRW dens.	MPRW homo.	Δ
ACM	0.77%	0.798	0.83%	0.838	+4.0 pp
ACM	1.46%	0.790	1.33%	0.828	+3.8 pp
ACM	2.67%	0.778	2.02%	0.811	+3.3 pp
DBLP	1.92%	0.727	1.45%	0.806	+7.9 pp
DBLP	3.23%	0.720	1.94%	0.786	+6.6 pp
DBLP	4.91%	0.698	2.49%	0.771	+7.3 pp

On ACM and DBLP, MPRW's retained edges have a 3.3–7.9 pp higher fraction of same-class endpoints than KMV's at matched density. The discrepancy is an unconfirmed empirical finding; a clean test of whether it is causally responsible for the CKA gap is an importance-reweighted MPRW baseline at matched density, listed as open work in §7.4.

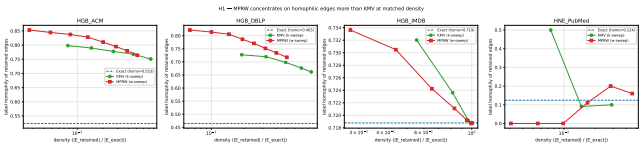

Figure 9: Edge-level label homophily of the retained adjacency vs. matched output density on all four datasets (Exact reference homophily annotated per panel: 0.523 ACM, 0.465 DBLP, 0.719 IMDB, 0.124 PubMed). On ACM and DBLP—where label coverage is full and the meta-path is not yet saturated—MPRW's curve sits above KMV's at every density. On IMDB the curves overlap. PubMed is shown only as a sanity reference.

Spectral radius. A second unconfirmed structural difference: at matched density, KMV's retained adjacency has $\lambda_{\max}$ that is 2–6× higher than MPRW's across all four datasets (Figure 10). We initially predicted the inverse direction (walks concentrating on hubs would yield high $\lambda_{\max}$); the data falsifies that prediction. The likely mechanism is that MPRW's retained edges form a tree-like walk skeleton with many leaves, while uniform-hash sampling preserves expander-like structure. We do not establish this mechanism beyond the visible inversion in the figure.

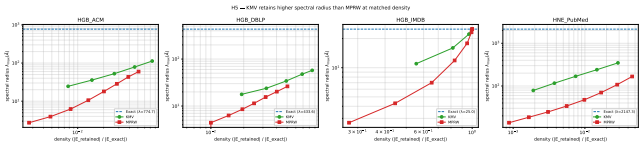

Figure 10: Spectral radius $\lambda_{\max}$ of the retained adjacency at matched density. KMV's $\lambda_{\max}$ is 2–6× higher than MPRW's across all four datasets at all measured densities.